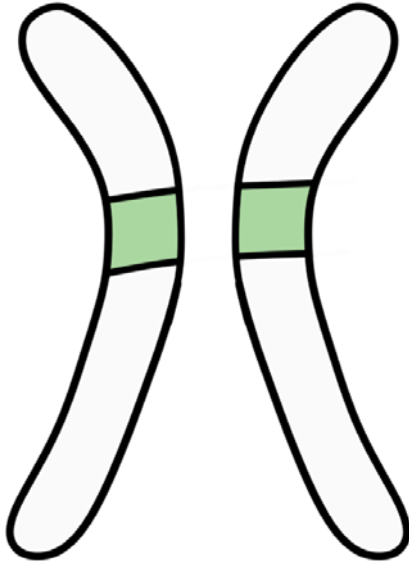


Amoeba Sisters Video Recap: Alleles and Genes

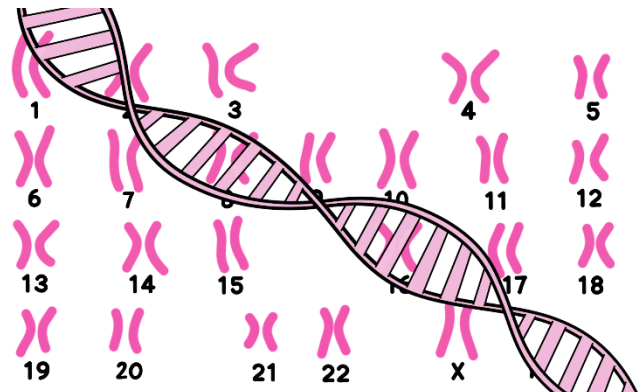
1. Visualizing the Vocabulary: For the following illustration, determine where you could label the following terms: **allele**, **gene locus**, and **chromosome**. Be sure to draw arrows to specify where you are labeling!



Analyzing Inheritance: A human's DNA code, found in nearly all body cells, can be condensed into **chromosomes**.

- How many chromosomes do humans have total in *each* *body cell? _____
- How many of those chromosomes in *each* human *body cell are from the mother? _____
- How many of those chromosomes in *each* human *body cell are from their father? _____
- How many *pairs* of chromosomes are there in *each* human *body cell? _____
- When looking at each pair, how many chromosomes in each pair come from the mother? _____ Father? _____

**some exceptions*



7. Working with the Vocabulary: *In your own words*, explain how a person may taste the bitterness of PTC by using the following vocabulary words in your explanation (choose any order): **trait**, **gene**, **genotype**, **phenotype**, and **alleles**. Underline each word as you use it in your explanation. In this explanation, you can treat PTC taste sensitivity as a single-gene trait. [As mentioned in the video, it may be more complex than a single-gene trait.]

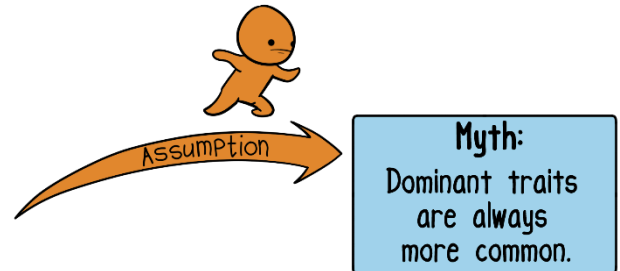


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8. **Deducing a Genotype:** If assuming PTC as a simple gene trait, what other genotype would you select to put in this missing genotype box that could result in this phenotype? Why?

PTC Taster (Dominant Trait)	
Phenotype	Genotype
	TT

9. **Relevant Scenario:** When explaining dominant and recessive traits to a younger family member, they respond, "Well chances are I can probably taste PTC, since dominant traits are more common." How might you address this misconception?



10. **Contrast:** Many times, students struggle with the difference between the terms **allele** and **gene**. How would you explain the difference of these two terms in a way that is memorable to you?

